

ActInf GuestStream 113.1 ~ Bayesian Mechanics of Economic Choice (Ernesto Moya-Albor et al.)

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Hello,
welcome.

It is June 26th, 2025

and we are live in active inference

guest stream 113.1.

Welcome Jack as well. We are with Sergio

Jacis and we will be discussing basing

mechanics of economic choice

computational foundations of economic

behavior. There will be a presentation

followed by a discussion. So thank you

again for joining and looking forward to

the presentation.

Thank you for joining Jack and thank you

for the active inference community for

inviting me to share my my current

research. Uh as Daniel said I am Samuel

or Sergio I have both names. Uh I am

residing in Mexico City. I am a

secondyear PhD candidate in artificial

intelligence and my current research

basically focuses on applying the active

inference framework to understand

economic choice and financial decisions.

Uh just to notice that this is joint

work with Ernesto Moya, PhD, my thesis

supervisor and Luis Alberto Casada, PhD

as well, which are two Mexican academics

as well. Well, let's begin.

So

basically I'm going to be speaking a lot

about this paper that we recently

published but I'm also going to give you

a brief introduction about the main

themes that we are trying to bring

together which are basically
topics from neuroeconomics topics
naturally from basin mechanics and the
active inference framework and topics
from classical or neoclassical
economics and psychology. So we're given
that we are trying to merge all of those
fields and as you probably know here in
the community the free energy principle
is very interdisciplinary itself. So I
think I would take some time to explain
the intellectual journey of these
different fields and how we think they
nicely come together. So we'll start
with economics. Then we would move to
how psychology meets economics. Then a
little bit of the thermodynamics of
information both for neuroeconomics and
active inference and then how the
divisive normalization model from
neuroeconomics can be derived formally
from the free energy principle. Uh so
that's the basic or the key insight from
our paper. And finally I'll I'll
share some of the future directions or
the future work that we intend to pro to
do for this year or the end of this
year. So let's start.

So I would like to start by
by sharing the let's say the
intellectual journey of how economist
usually think about valuation and about
encoding value. So let's start with the
Swiss mathematician uh Daniel Bernuli
and as you probably know he's very well
known for creating uh classical utility
theory or expect or classical expected
utility theory which basically said that

that economic agents or people don't encode does don't encode absolute monetary values. We don't encode absolute monetary values. we encode what he called moral value and moral value was just a way of saying subjective value. So monetary values are are I mean are subjective on how much we want something on the time we are on the on the place we are. So basically that's what he formalized and he formalized it as a weighted sum of expected probabilities and payoffs. Now later on very famously von Neuman and Morgenstern published their famous work theory of games and economic behavior where many of you listening uh probably have heard it because it that was the the foundation of game theory but one key insight and was that when they published this work they just expanded Berni's expected utility theory to account for the specific axioms that economic agents should abide to in order to be what we usually came to to refer to as rational to make rational decisions or rational rational choices. So these uh these axioms were that our decisions would be should be complete they should be transitive and this this is very very important. So when I prefer A to B and B to C uh we should follow this this this ordinal uh property and our choices should be value independent independent of the other choices independent of past choices. So that was another another choice and if we if our rational agent or our economic agent abides by these

axioms the key point here is that what naturally follows and can be derived mathematically is that this agent would be maximizing its expected utility. And this is key because this is another way for example in the active inference uh slang or language that this economic agent would be pursuing a goal. So it's another way of saying it it's going to be focused on goal directedness. So so just keep that in mind and we will see how that impacts us later. So this this framework uh basically impacted all of neoclassical economics and modern uh economic development. And another key aspect of Bonoyman and Morgansson's theory is that they thought of the probabilities of events and the payoffs as objective. So what I mean what what do we refer with that? For example, if we are going to roll a die, we know that we know the exact probabilities and we can compute them exactly analytically. So that's pretty much what they what they meant that we know the the objective probabilities of specific economic outcomes and we know the payoffs in advance. However, naturally you can start thinking that that's rarely the case with the exception of for example uh specific uh random games or casino like games. later on Sapul Samuelson uh which probably one of the let's say leading figures in the in in economics in your case in economics uh came to say that we really don't know the preferences of economic agents or economic agents uh I mean we cannot get

inside their brains to know their preferences so in their in his paper consumption theory in terms of revealed preference he just say he basically said we cannot know the preferences in advance But we can observe their choices and infer their preferences. So it was that was that was the first type of a bit of Bayesian approach to economic decisionm. But later on Leonard Leonard Savage or better known as Jimmy Savage uh famously published the foundations of statistics in the 1950s where he established subjective expected utility theory meaning that he established that we definitely don't know the preferences in advance uh and we don't know the objective probabilities. The only thing we have is beliefs. Beliefs like in the the active inference framework. So those beliefs are subjective probabilities. However, one important aspect here is that even though he established that we will have these beliefs and these objective probabilities, this would be a normative theory. So normative theory is a way for economist to say rational because many many people not coming not acquainted with a theory of economics uh hear rational and they just think of I don't know like capital spuck or something like that. So when we say rational we really say normative and what we refer to is that the economic agent is making logically consistent choices abiding by these axioms transitivity independence uh completeness and we can also add

continuity.

All right so that's basically how rational choice theory evolved from its beginning and pretty much up to nowadays.

Now then

many many other researchers and economists came to say all right so that's the theory but let's observe how do people actually behave so let's build a descriptive theory not an not a normative theory of how people should behave uh and the first uh key figure in this story it's Maurice Allais uh which was a French economist uh he also was another laureate in economics and he came up what later became known for uh as the the Allais paradox. So the Allais paradox basically said that

many sophisticated decision makers including leading economists like Jimmy Savage consistently violated the independence axiom that we just referred to. So so what what did that mean?

That basically meant that when facing different choices and rearranging the the many of choices or the opportunity set of choices even Nobel laureate economist didn't abide by the the independence axiom. So that was one of the first uh let's say empirical validations that that the axioms of rational choice theory were consistently violated even by the people that came up or that that developed the theory for them.

Later on and most famously Kahneman introduced uh psychology into economics. So that was the point where psychology

formally met economics and they they came up with a formal descriptive theory which was later called prospect theory in the 19 1980s. Uh so basically prospect theory was an empirical theory contrary to the previous theories which were mainly theoretical and they literally observed how people make economic decisions under uncertainty and they come came up with a graph here that we have in the right hand side corner which you can see that has a sigmoid function. So let's contrast that function with the classic utility theory function from Brendle Lee. So you can see that the classic utility function from Brendle Lee is basically explaining what we usually learn about in economics 101 which is diminishing marginal returns. Uh we experience diminishing marginal returns. That's the basic the basic utility function. But what Cananan and Tverki find out was that that was not how people behave in reality. They observed that they behaved like this uh with this utility function which is this sigmoid shaped utility function. And the key aspect of this utility function is that there's a discontinuity between gains and losses. they are they more or less found that uh people or the brain registers with twice the intensity losses to gains. So that was a key key insight and a key violation of traditional rational choice axioms. Uh so we can speak about a discontinuity between gains and losses. uh and

popularly this theory came to be known as the let's say irrational argument for economic behavior or economic decision-making. Uh but the the thing was that many economists push back against this and many other decision scientists push back against this because this theory essentially said people take behave and take irrational decisions very often. they are subject to biases. They take a lot of their decisions with heuristics that make them fall into traps. So essentially he was saying that people people are not very smart. People are idiots even. So that's what they said. Now later on uh Richard Taylor and Loewenstein and other relevant figures who will also would be well in the case of Richard Taylor was also awarded the nobility in economics uh pretty much imported prospect theory formally into economics and establish what we now know as a field of behavioral economics which is accounting for these uh biases and heuristics into economic decision-making and pretty much checking how consistently these biases heuristics make us violate the traditional actions of neoclassical economics. However, however they I mean everything was like Paul Samuelson said it we infer the preferences we infer what is happening in the brain by observing behavior. It was still not uh a neurophysiological theory or anything like that. So that was pretty much how we have been uh evolving. Now let's move let's switch

breaks and let's move to some other things happening specifically in computational neuroscience. So in computational neuroscience more or less in the same along the same time uh there were a lot of very interesting discoveries. So famously, Simon Laughlin's 1981 study of fly photo receptor neurons demonstrated that the optimal computation under constraints under energy constraints would be one that look like this where where the fly or in this case that the the flies neurons photo receptors would spend more on representational resources uh on the parts that were more common or more likely to appear. Meaning that for example for certain light intensity that was more common to appear in in its environment the fly or the the flight photo receptor neurons would encode those would give a higher probability of encoding that level of light more often than very rare events in its environment. So basically what what we are seeing is that the fly photo receptors are mapping the environmental statistics in order to optimize or to compress the amount of limited energy it has in its brain. So as we know uh neural computation is very expensive. The brain even though it's it's only more or less 2% of the body's weight it uses about a fifth of the body's energy. So, so the so naturally that means that it's very metabolically costly and it needs to be very very efficient on using that energy for its rep representational

capacity and also we can bring uh evolution and adaptiveness to the to the mix because naturally the fly as long as it performs this type of action and other type of actions better uh is going to be able to survive. uh I mean it just is just going to be able to adapt uh more more optimally. Then we we can also introduce Landau's principle from Rolf Landau who said decreasing stochasticity in choices decreases information entropy and later the in the famous infomax principle which says the brain optimizes mutual information between output and input. So that basically said we have we have an example for the fly the fly has an unlimited number of pixels that or the of just stimuli that it has to encode but it has a very limited input capacity. So it has to compress that and it compresses that by assigning different weights and assigning higher probabilities to the in this case the light intensities that that came that came up more often. And when when we take the integral of that function, we can see that we're going to see this sigmoidal this sigmoidal uh shape or the sigmoidal uh CDF. So just remember that because that's going to come up later. All right. So how can we take these these insights and bring them back to economic to economic decision making? So if we assume what rational choice theory said to us that we essentially make uniform uh we make decisions with uniform precision we don't have information asymmetries as new classical

economics said then what that means in terms of information processing is that neoclassical economies like bonoyman and morgan assume that there was no cost in representation.

So and we can translate that uh geometrically as uniform precision and you can see as you can see in the graph.

So without depending on the reward magnitude the fine rate is going to increase just linearly. However the reality is very different. The reality is that if we put some noise into the mix and the noise is represented by that uh gray area you can see that the rewarded magnitude by adding just a little noise is going to be huge. It's going to be huge. So it can be 20 or it can be 40. So it's huge. So what this means in terms of Laughlin's uh uh fly experiment is that assuming that this was the brain how the the brain was encoding value, the brain would be very wasteful on encoding all of the possible rewards at the same at the same rate or at the same probability. So that would be very wasteful in terms of energy for the brain. So the reality as we came to know is that that that the optimal way of allocating uh energy resources for the brain is to allocate them for those events that came up more likely. So that's a key aspect. Now another important thing is that this also tells us that the violations commonly discovered by Allay Cannaman Traverski and other behavioral economists perhaps weren't actually violations to rational

choice theory perhaps rational choice theory was wrong and in reality those decisions are in fact optimal are in fact normative but normative under the assumptions that we have thermodynamic constraints that we have a limited representational capacity So if we assume that we have those limits on representation then those decisions become rational.

All right. So then how should value be represented in the brain? Let's remember from Barlo's efficient coding hypothesis narrow neural systems involved to maximize information transmission while minimizing metabolic cost. So let's see the sigmoidal curve in the in the panel C. So that's basically the sigmoid function that we just showed you. And notice that it's remarkably similar to the value function that Kenneth and Tverki discovered through the behavioral experiments the that that the the very well-known loss aversion sigmoidal function. And also notice that the optimal precision allocation produced a curve that is reference dependent that is a reference dependent uh response function. Um so basically what what this means is that what behavioral economists interpret as cognitive biases might actually be uh an optimal representation or an optimal computation of the brain of value under biological constraints. So basically we can see that notice the what I just told you the gray area if we assume uniform precision in the lower right hand side corner in B the the

noise made that the reward magnitude was huge. So the amount of energy expenditure is going to be huge. However, if we assume the sigmoidal function which uh that would be the function that we would get as we saw with llings under thermodynamic constraints that noise is very small that the the reward magnitude is very small. So that's basically the the evidence or the yeah I mean the demonstration that the brain is very efficient what when using that sigmoidal shape instead of using that uniform shape.

All right. Now moving to neuroeconomics. Basically neuroeconomics uh arise mainly from combining neuroscience, psychology and economics. And one of the leading figures I'm going to tell you about uh well some of the leading figures uh let's uh let's start with uh David Hager. David Hager which was if I remember correctly an NYU came up with uh with other authors like Simon Chel and Karandini with famously divisive normalization model. So what is divisive normalization? So device normalization is basically uh a computational algorithm that the brain uses to well originally it was it was used to explain how the brain or how the the V1 uh primary visual cortex encoded visual information. So basically if we look into the graph what what Hager and colleagues pretty much found is that neuro I mean neurons in the retina basically encode information but they

don't encode it as we saw with Laughlin equally or uniformly. What they do is that they divide the specific input that or that the excite the excitatory input that we want to see it divides them by the pool of surrounding neurons. So, so, so by dividing them by that that pool of surrounding activity of of neurons, what it's basically doing is it's subtracting it's taking out that redundant information. So that's pretty much what it's doing when dividing it by that by those surrounding neurons. So that makes the again that makes the the brain or in this case the the neurons in the retina much more effective or much more efficient. And if we see how do the the the value encoding looks like. So we can see that the graph in the right looks pretty much the same as Laughlin's uh graph for fly photo receptors. Later on uh other neuroeconomists came to see that that divisive normalization algorithm not only worked for accounting for information in the primary visual cortex, it also worked for all information for sound and also for encoding value in the posterior parital cortex. So basically uh what later came came to be known is that it appears that device normalization is a canonical narrow neural computation and not only for vision and later on uh Louie Kenlie showed that value representations in the brain can be well described vivasive normalizations. Also by this time uh neuroeconomies have found have founded the the main brain areas that that are

probably responsible for the main activities of encoding value specifically the ventromedial prefrontal cortex and the striatum. So even though there are other brain areas that also play a role for example the amygdala lines of cases and all like the ventral prefrontal cortex and the striatum are definitely have been shown to be the key areas involving value representation in the brain. Now later on Paul Glimcher uh here which is also a professor at NYU. He actually directs the neural institute at NYU. Uh came to I mean just keep keep kept working on this work with also with David Hager and pretty much imported device of normalization from neuroscience from vision neuroscience into economics and later uh with joint work with Timura which is now now I think in in Australia or in Israel uh later uh they they published their work on expected subjective value theory. Now notice that subjective value theory looks a lot like Savage uh subjective value theory but this is a narrow science-based subjective value theory. So even though it's it looks a little the same is not the same. So basically they he established he took this device normalization model and established that uh and he used this formula to say that the brain can basically well that value representation in the brain can be shown to follow this exact formula. So what this formula is saying is subjective value SD is dependent on

the excitatory value the value that I'm trying to evaluate or the payoff divided by a fixed constant plus the payoff what we call the payoff expectation which is the normalization factor which in the in the case of vision neuroscience what the sum was the sum activity of the pool of neurons but in an economic interpretation it could be the alternative options options. So if we have a menu of economic options, it will be the the the average sum of those options. If we are speaking about intertemporal choice, it could be the the average uh the weighted average of those payoffs across time. So basically again what we are saying is is that the value representation in the brain is context dependent. So what what what this model is basically bringing to economics is context modulation context modulation in economic decision making in the brain and I mean just for you to know in the standard devices normalization model here use usually FR instead of ST to represent that ST can be interpreted as neuronal firing so it's literally a neurophysiological model uh for economic value representation

All right.

Now, moving on.

If we dig deeper into the model, we can see that it has two key parameters. It has alpha and m. So, what alpha is uh in the in this type of model, it's considered a free parameter. But in a in an economic context, it can be

considered for example the risk tolerance. its because that parameter alpha controls the shape or the curvature of the of the function. So notice that if we increase alpha we get pretty much in the in in panel A we get B's utility function the same function that Daniel Verni discovered like 200 years ago if we increase alpha we get that that type of shape and then if we decrease it a bit we get canverki sigmoidal function so the key aspect here is that this model is a more general model because it allows us to account for different utility functions that had come up in both neocclassical economics and behavioral economics. But again this is a neuroscientific physiological representation uh it just happens to look like neocclassical and behavioral economics and also form which again is the pay of expectation uh what it does is that that it moves the the mean the center of the of the of the utility function. So we can move it by increasing m. And what that means in terms of basian of of basian decision making is that depending on my experience on past outcomes on my past preferences that's going to shape that's going to move the shape or the the center point of my objective value uh utility function. All right.

Um

yep.

Now moving on into act the active inference framework. uh I mean this is

going to be uh pretty got pretty familiar to many people here listen listening to the talk in the community but uh for those maybe not so well acquainted with uh the free energy principle uh we can trace back let's say the yeah the like the intellectual origins of the free energy principle to hemholds who basically uh discovered that that we have as we have as individual unconscious inference And this might might sound very similar to what Paul Samson said that we cannot look inside brains to see what preference people do do people have. That was pretty much the the same type of things. But he was speaking about neurophysiological processes. Later on, Hinton and Moonfort uh came up with the first energy based models uh which later became very famous for example with the Bolsman machine uh and those were the very the very type of uh the very first models that were energy based and that considered hidden or latent variables that so that's a key a key aspect and specifically Moon for came up with the basin approach to vision later on predictive coding with row and Balard uh pretty much uh for predictive processing uh lay laid out the the ground for for later on professor Fristston to came up with a free energy principle and it's let's say practical framework which is the active inference framework and later on with basin mechanics. Uh so the key aspect here is that we can regard economic choices essentially as

predictions about future act outcomes or beliefs about states under uncertainty because even if we are going to make an economic decision in the present we are going to see we're going to think about future outcomes about how much would will like this chocolate this coffee what if I spend my salary in this or that how is going to impact that on my finances. So almost all economic choices naturally inevitably have to do about have to do with the future and have some uncertainty embedded in them.

Now

for those who are who maybe are not acquainted with the free energy principle but for those who are uh I think it's never uh it's never wasteful to go again to explain it. So basically that what the free energy principle claims or says is that if we observe that things exist if if something persists over time and its identity can be re identifiable in the environment. It has a given structure then it must be true that it must encode or instantiate a statistical model of its environment or we can call it a generative model an internal model of its environment. There we have the famous uh free energy principle formula and the key aspect here is that the free energy principle has two aspects. It has an accuracy term which basically that could be uh rendered as we're going to see later as the expected utility in in economics expected utility theory the log evidence and that's basically how

accurate are my my predictions. I wanted to have the the most accurate predictions. So that's basically the like reward in terms of reinforcement learning and in terms of the complexity term the complexity terms brings into the model uh an entropic an entropic term. So what does entropy bring into the model? It brings flexibility. It it brings flex flexibility for for example in e in the economic context flexibility for preferences. We have to we want to be flexible in our preference because we have uncertainty in the world. So perhaps now I like something but later I will not like that and I would prefer uh X to Y. So basically the model can account for uh for biases that that were regarded in behavioral economics like like intritive decisions uh or just flexibility in in preferences. Now the question would be does the expected values theory that uses the devising normalization model from their economics can emerge from the perceptual inference uh part of the free energy principle. So that's the question that we are asking and why specifically the perceptual inference in the perception action cycle of the free energy principle because even though active inference is well the active part or the action the action part of the cycle is very important because when we have references we have preferred outcomes we're going to be looking for information we're going to engage in information gathering activities. However, the initial value

representation in the brain is going to be perceptual. It's going to be perception. So the so we can say that the if the model exists is going to be living in the perceptual inference section and later on we can add the information gathering aspect uh through the the action component in the model. So as we mentioned what active inference does is that it assumes that agents and in this case economic agents are not passive observers. They are going to look to fulfill their preferences their outcomes and to have better information to make better economic choices. Now as we just saw if we take out the complexity or the entropic term from the free energy principle what we get is just the expected log probability of our preferred outcomes that can be more or less approximated by the expected utility in economics. So that formula that expected utility in economics was basically what Bonoyman and Morgansen uh formulated uh almost 100 years ago. So basically in if we think about this model in an economic context what we call preferences in economics can be regarded as prior beliefs or preference distributions under the free energy framework. Now what is that? I mean how is active inference useful for economics? Well, the first thing is that it can unify the epistemic value or the information gathering aspect of of economic behavior or strategic behavior and decision making. Um, also as as I just mentioned epistemic versus pragmat

pragmatic actions. So I mean economic agents uh and strategic agents need to evaluate the cost of gathering information versus increasing their precision. So again as we mentioned uh having unlimited precision is not viable having having more uh well less stochastic or less noisy decisions is costly. So we need to evaluate the cost or just just weight the cost of those decisions. And finally, as we also know, there's a lot of research coming out in the on the free energy principle and active inference applications to artificially intelligent systems that this framework allows us to track back uh all of the specific decisions and all of the specific preferences and pretty much all of the parameters in the model. So, contrary to other machine learning techniques like neuronet networks, this is not a black box. This is a transparent model that that allows us to trace back every decision made in the model to the specific parameters of the model. So it's a very transparent and explainable uh decisions uh decision system if implemented specifically in an in an in an AI system.

All right. So moving on to the key uh key let's say contribution of the paper. So what we just we just uh we showed is that if we if you start from the free energy principle and the expected free energy, you can derive from that the devices normalization model as assumed by the expected subjective value theory of Glenmer and Timola. And basically if

you do all of the math you can get to the free energy minimization and get M like the the internal model or the the mean under perceptual inference as that formula. And we can see in the in the lower lower uh box that that formula and each component of the formula can be mapped directly to the subjective uh value expected subjective value theory or the divisive normalization model. So what the lemma says is under a hierarchical generative model with gauss and priors and likelihoods perceptual inference through the free energy principle converges to a representation which is that representation there which is equivalent to divisive normalization. So what is going on? As we as we said the uppers upper part of the of the model is basically the predicted value the exact the excitatory value that we are trying to encode and we are dividing it by a baseline plus uh weighted average which is in this case is a precision weighted prediction error minimization term. And again what we are doing is we are subtracting we are taking out all of the irre all of the irrelevant information that we already know. We already know from our experience from past outcomes or or we just don't care it's irrelevant information or so we are taking it out by making that division in order to just focus on the specific value that we are trying to encode. So in short what we are doing is that we can show that the free energy principles allows us to

account for context dependent economic decisions and also reference dependent economic decisions as posited by behavioral economic theory. But in this case again is not a bias in is not a heuristics per se. It's actually an optimal decision. It's a normative model. So in the case of uh of the economic uh let's say language this is a rational model of economic decision- making but it's rational by assuming thermodynamic constraints. Uh so that in that sense it becomes rational. So we're really we're really excited about it about this because what it says is that from neuroscience we can derive some of the let's say of the the the oldest paradigms in economic theory but we can test them and validate them both mathematically in this case but also empirically because I didn't mention that but the divisive normalization algorithm has been tested empirically in humans and in non-human animals as well uh from I don't know like monkeys and I think even bees. So basically we can see that the device normalization model is a very robust model and as many people in the community here in the active community know the free energy principle as well can be mapped into a narrow process theory. So moving into that basically the future directions of our work is that we want to operationalize this work into a into the let's say op into the practical version of the active inference framework which is in in into

a graph into a partially observable
markup decision process. So basically
what we want to build is an agent uh an
agent that makes economic or financial
decisions in the market. And how is this
modeled better than for example uh a
transformer type of architecture that
does this type of economic decision
making. Number one it's the traceability
as we just mentioned. uh when we when we
use transformer like architectures or
neural network architectures for things
like financial decision making or things
like health uh decision making it's very
sensible because we don't really have we
we cannot really account for what the
model is doing or how the model is
coming up with its output in this case
with this type of graph uh in focus we
can have traceability of the system
number two everything is explicit. All
of the parameters are explicit. Uh so
that's another thing we can for example
uh make beta and and the rest of and
gamma and the rest of the parameters
very very explicit. For example, we can
uh use gamma as investor confidence. We
can use beta as decision uncertainty in
the model. And finally it can uh we can
use the the basian update uh belief uh
equations to account for adaptive
learning because naturally when you
become an investor or for any other type
of economic decision setting you learn
as we just mentioned you learn about
past outcomes if you have bad outcomes
while investing you usually become more
conservative. Uh your risk tolerant

changes your priority and and financial objectives change with time. So, so basically this model also allows us to have that flexibility uh for uh for for how the agent is adapting in the environment. So, basically this is a a future work that we are that we're currently building and we hope to have empiric empiric uh results perhaps later this year or the the beginning of 2026. So, this is pretty much the the future we're working on. Um, and yeah, and that's pretty much the end of of the talk. Uh, we'll open up the the space for questions. Thank you, Daniel. And here's my my contact and my LinkedIn as well. If anybody listening wants to add me or contact me, uh, I'm all ears. Thank you, Daniel. And thank you to all all for listening.

Thank you. Very cool.

All right.

Well, those watching live can write a question in the live chat. Let us begin with Jack. Would you like to say hello and give any initial reflection or question?

Hey there. Yeah, I mean I really find it fascinating that what we might consider biases in a lot of economics kind of turn out to be optimal. Um, you know, one thing I I want to understand more is why is it important that you can derive some of these up-to-date economics models from active inference? Like what does that say that they are so related? Um, and um,

is it natural to think that they would be like when you set out? you think maybe like maybe this won't work like tell me bit a bit more about the fact that they are so related. Yeah. Well, I think that um a long-standing inspiration in the economics field is to make economics as scientific as possible within the social sciences. So historically economics imported a lot of things from physics um like I don't know like thermodynamic equilibrium and many other things uh just precisely like like like trying to trying to make it as scientific as possible but what happened is that it became highly technical uh and highly mathematical but always assuming that that economic agents behave as if so theoretically. So just like like a highly theoretical field. So what happened initially with behavioral economics and I think can happen now with neuroeconomics and by importing uh tools like the free energy principle or methods is that we can make economic theory where we can validate or we can test economic theory in real life experiments uh or or in or in my case also mathematically but mathematically like the Pythagorean theorem. I mean even we can demo even even though we can demonstrate the py the pythagoran theorem mathematically we can also test it in reality. What happened with economic theories is that we prove them mathematically assuming certain very uh implausible axioms and later we just

said well it's demonstrated but assuming those actions. So what this allows us to do is to prove it mathematically as well but by accounting for a in this case for a narrow process theory that can be testable in uh in empirically. So that I think that that's a key a key insight and in this case a principle uh I mean if we as myself and I think that many people listening in the community assume I I don't know I mean I don't want to use the the word true but it's a very good model to explain how uh to explain self-organization and how things exist. If we think that that's a very useful model then it should apply to this as well. So that I think that's another thing on on the second part of the question.

Thank you. Well, I'll ask a question then Jack you can you can go again or I'll read one from the live chat. So how does this help us think about the microeconomic macroeconomic relationship like what what do we get from this cognitive approach or from basian statistical approaches that might connect those two differently than how they're understood to be connected now. Yeah. I mean, a recent trend that that that I think that has been going on in in the last decade perhaps in in just standard economics is that there has been u let's say uh I mean let's just say that that macroeconomics has has decreasing popularity. So for example, famously MIT is no longer uh requiring

to take macroeconomics as a mandatory course even within uh within the economics major. So that tells you that that in a sense uh we are migrating uh to yeah I mean ju just to explain economic in the individual level and then try to go up to add them up. So I mean at least from my perspective the future of macroeconomics and again that's my own my personal opinion would be starting with these type of empirically grounded microeconomic models. Uh so we can regard these type of approaches like a a like a microeconomic model and then moving it up in in a microeconomic setting but but relying on complex systems in in complex adaptive systems. uh so in that sense again I think that the energy principle uh blends very nicely with that type of approach.

So that would be my my opinion on that.

Yeah. Go ahead Jack.

Maybe I have a more general question and it might not be answerable by you, but I wonder why biology would be so optimal.

Um I guess um because you know when economists do experimental research and they find oh maybe people have biases and stuff maybe

maybe that makes sense like why would we be optimal but then kind of what you're saying is well we are optimal given the constraints that we have the fact that we're biological and we don't have infinite energy and things like this.

So, um I don't know if that's a question, but more a comment of um I

how much resources are being obtained per unit time. Then once you start to bring in different trade-offs like water loss or the costs of movement, you start to get closer and closer to empiricism. And then the sort of um death blow to that just seeds per minute uh measurement is the subjectivity of beliefs like well there's predation risk. So you could ask is this G behavior optimal given predation risk? But then in a way when you get to the subjectivity of predation risk that's like kind of asking well is this an optimal invest investment strategy given the volatility profile and it it may or may not be. There may be a better performing trade strategy given a gi market condition. But in a way once you get to subjectivity which is the sort of nucleus of decision-m you get to that tautology which is like well they're acting as if these are the conditions. So given their subjective evaluation they are acting in this baze optimal way. And so in a way that sort of um like fake out almost like oh it's rational but that's so much of the debate that at least I've heard from afar is this debate what is rationality are are people in their decision-m rational or not and it's like when we're doing baze optimal modeling even if it's going to be a pathological state it's like there's something about the parameterization

where it is optimal for this repetitive behavior to arise or is it's not that everyone around this person or the person themselves wants to be having this thought or action per se but optimal doesn't mean that it means like the ball is at the bottom of the bowl but the bowl shape so it's like the starting point for basic optimal modeling is that by taking constraints into account including interior constraints, subjective constraints and evaluations, then it's a logical and not really an account to say that it's optimal. It's just sort of things are how they are and all the work goes into structuring and doing the inference on what those constraints are.

And so it's like the framework says less about what's optimal, but it lets us look at traders with a variety of portfolios and say we're already going to take it as a granted that they are acting in their own best optimal way. So what is the structure and the parameterization that enables that and in that way feels like it moves forward.

Not trying to imagine a time before there was this debate of rational or irrational but but just as you structured the presentation moving to a place where we can empirically ask what are those objective and subjective constraints that make decision-makers act the way they act.

you're in the AI program. So just how

did that like come to be and did you how
did you come to this economics
topic in that program? Right. Right. Um
basically my background uh originally is
in finance or let's say financial
economics. I later pursued uh graduate
studies in engineering mathematics.
uh and while at that I I did my my
thesis on complex adapt adaptive systems
or complex systems engineering so and
that inevitably or eventually led me to
active influence of the free energy
principle u and at the same time well
when I became very excited about that
like I don't know like three four years
ago uh I also pursued the graduate
studies in uh in psychology and and
focused on mathematical psychology so
that was like the natural bridge for for
me to go into free the free energy
principle. So later when thinking about
what field I wanted to pursue for my
PhD, I thought well I mean usually
economists are very I don't know I don't
know why but they usually work like a
lot in silos. I mean they they work on
things like I don't know like law and
political science and so sociology but
not so much in other areas like I don't
know like not not not so much in in
technology and stuff like that usually.
Uh also in finance I mean they just
don't work in these type of things. Uh,
and in psychology, I mean, we I didn't
have a I mean, at least here in Mexico,
we didn't have we we I don't think we
have a very strong mathematical
psychology program. So, the only place I

thought I could merge
decision sciences, cognitive science and
and my interest in economics was in AI.
And given the the the recent literature
or I mean I don't know the last five
years the literature that has been
coming up about applying the active
inference framework for practical
applications uh I don't know like
inverses and stuff like that. I just
thought that that was like the perfect
route for

so that that's pretty much the story.

Which economic actors or firms
would have a different day-to-day
experience? How if these kinds of
relationships were better understood?

Well, I think that for example on the
type of agent I am uh I intend to
develop.

one one reason why I mean I don't know
if you you have a different opinion but
uh I haven't seen many AI agents for
economic advisory I mean there's some
there's some there's some robo advisors
but they're very let's say very simple
uh and again the reason why they're
they're very simple I think is that it
has been proven and it has been tested
that LLMs cannot plan so for example The
other day I was listening to a talk from
professor Carlo Frison and he said that
at least for him an agent for it to be
an agent needs to be able to plan. So I
I completely share that insight. So if
he if it could mean if it has been
proven that LLMs cannot plan then at
least for me they are not very useful

for financial advisory because the key for financial advisory is financial planning. So, so in that sense I think that looking for other type of architectures in this case the partially observable market decision process that that is proposed by the active inference framework could be better for that I mean just for the fact that it can plan and second because it's explainable and as I was saying a key aspect also about or the a key disadvantage of using transformer-like architecture for uh some sort of financial advisory is that it's not explainable. So if somebody loses all of their savings, so what's going to be the explanation? Uh just hallucinated or or something like that. So I think that that's a practical application.

There will be some very interesting economic arenas and futures with all these different kinds of agents making different belief updating and policy selection. Um yeah policy election as well.

Jack, do you have anything else you want to ask?

No, not really. Thank you. It was really interesting. So, what else will you do in your PhD?

Hypothetically, but what what what what where would you like to take in the second half of the PhD?

Well, basically

uh finish building this this agent and hopefully get to test it uh like in with real people. Uh so that would be my my

intent for the next couple of years.

Anything else you want to add?

Uh well just again if somebody wants to get in contact for I don't know any type of research opportunity collaboration or anything there's my email my academic email and my LinkedIn profile as well.

So so happy to collaborate. Cool. Well well thank you again for the very presentation and work. Absolutely.

Thanks a lot. Okay. Till next time. See you. Thank you. Bye.